

CHAPTER 8 : FREQUENCY CONVERSION

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Mixers : electrical circuits that convert frequency

In TRC-II :

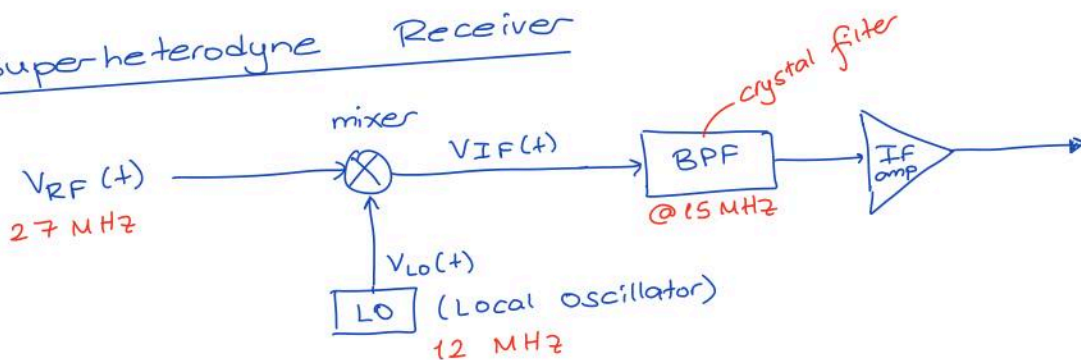
speech/music signal
(~3 kHz BW,
centered around DC)

27 MHz for transmission/reception
(amateur band)

15 MHz (IF frequency)
for filtering

shift to higher freq. → up conversion
shift to lower freq. → down conversion

Superheterodyne Receiver



$$V_{IF}(t) = V_{RF}(t) \times V_{LO}(t) \quad \text{where } V_{LO}(t) = V_{LO} \cdot \cos(\omega_{LO} t)$$

* if $V_{RF}(t) = V_{RF} \cdot \cos(\omega_{RF} t)$

$$\begin{aligned} V_{IF}(t) &= V_{RF} \cdot V_{LO} \cdot \cos(\omega_{RF} t) \cdot \cos(\omega_{LO} t) \\ &= \frac{V_{RF} \cdot V_{LO}}{2} \cdot \left[\cos(\underbrace{(\omega_{RF} + \omega_{LO}) t}_{@ 39 \text{ MHz}}) + \cos(\underbrace{(\omega_{RF} - \omega_{LO}) t}_{@ 15 \text{ MHz}}) \right] \end{aligned}$$

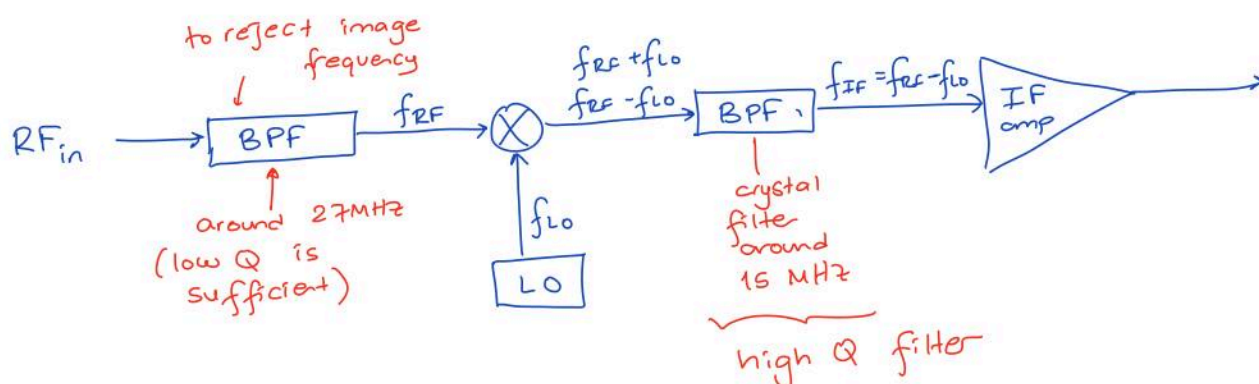
* Crystal BPF centered around $\omega_{IF} = \omega_{RF} - \omega_{LO}$
filters out $(\omega_{RF} + \omega_{LO})$ term.
So, only the signal with $\omega_{RF} = \omega_{IF} + \omega_{LO}$ is selected.

* However, there is one more frequency that can pass through the BPF

$$\text{if } \omega_{RF,i} = \omega_{IF} - \omega_{LO} \quad @ 3 \text{ MHz for TRC-11}$$

↑
"image frequency"

$\omega_{RF,i}$ is $2\omega_{LO}$ lower than the desired RF.

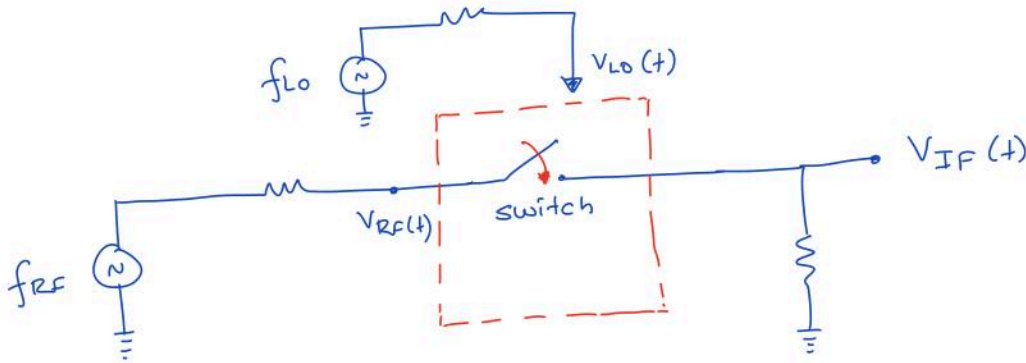


	<u>RF freq</u>	<u>IF freq</u>	<u>LO freq</u>	<u>Image Freq</u>
AM radio	535 - 1700 kHz	455 kHz	990 - 2155 kHz	1445 - 2610 kHz
FM radio	88 - 108 MHz	10.7 MHz	77.3 - 97.3 MHz	66.6 - 86.6 MHz

Mixers

- Nonlinear devices with 3 ports to perform multiplication
- Built from diodes, transistors, or other non-linear devices.

Switch mixer: (to understand the concept)

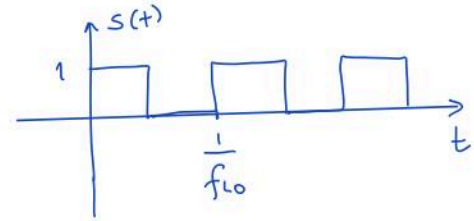


Assume $V_{RF}(t) = A \sin(\omega_{RF} t)$
 $V_{LO}(t) = \sin(\omega_{LO} t)$

* Switch is opened and closed based on $V_{LO}(t)$.

$$V_{IF}(t) = V_{RF}(t) \cdot s(t)$$

where $s(t) = \begin{cases} 1, & \text{if } V_{LO}(t) \geq 0 \\ 0, & \text{if } V_{LO} < 0 \end{cases}$



* $s(t) = \frac{1}{2} + \frac{2}{\pi} \sin(\omega_{LO} t) + \frac{2}{3\pi} \sin(3\omega_{LO} t) + \frac{2}{5\pi} \sin(5\omega_{LO} t) + \dots$

* $V_{IF}(t) = V_{RF}(t) \cdot s(t)$

$$= \frac{A}{2} \sin(\omega_{RF} t) + \frac{A}{\pi} \left[\cos((\omega_{LO} - \omega_{RF})t) - \cos((\omega_{LO} + \omega_{RF})t) \right]$$

$$+ \frac{A}{3\pi} \left[\cos((3\omega_{LO} - \omega_{RF})t) - \cos((3\omega_{LO} + \omega_{RF})t) \right] + \dots$$

So, IF output of the mixer contains sinusoids at many different frequencies:

ω_{RF} , $|\omega_{LO} - \omega_{RF}|$, $(\omega_{LO} + \omega_{RF})$, $|3\omega_{LO} - \omega_{RF}|$, $(3\omega_{LO} + \omega_{RF})$, $\dots$
 @ 27 MHz, @ 15 MHz, @ 39 MHz, $\dots$

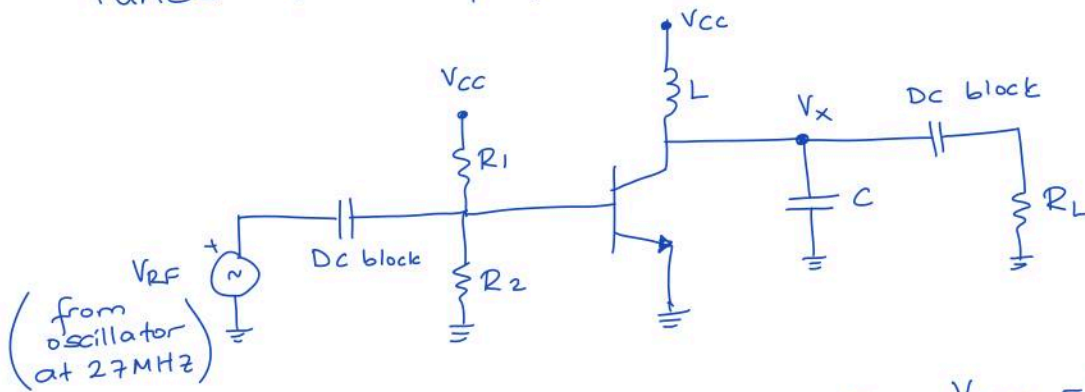
Crystal BPF will select this component.

Conversion Gain of a mixer :

$$G_c = \frac{P_o}{P_{in}}$$

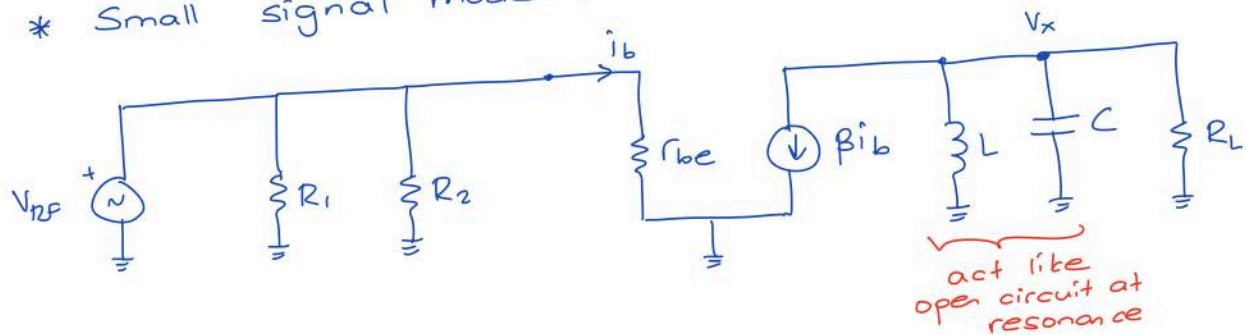
Amplitude modulator : (up conversion)

Tuned RF amplifier with its supply voltage modulated



L & C are resonant at 27 MHz. $V_{x,DC} = V_{CC}$ (L is short)

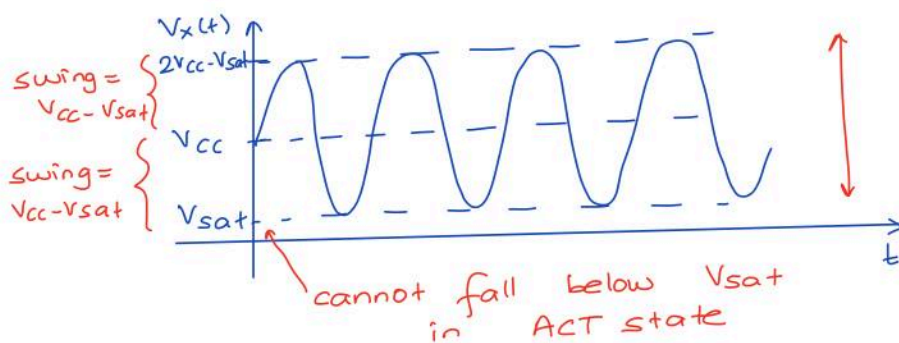
* Small signal model :



act like open circuit at resonance

$$V_{x,AC} = -\beta i_b \cdot R_L \quad (\text{at resonance})$$

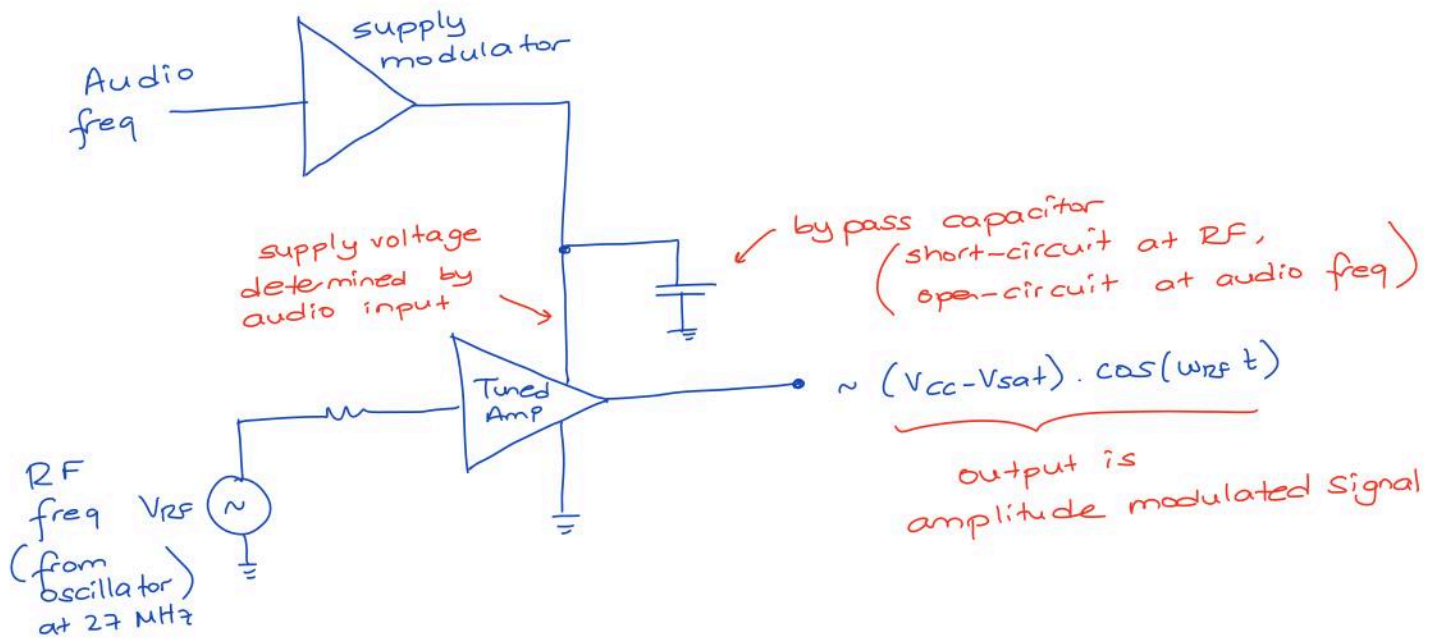
$$V_{x,AC} = -\beta \cdot \frac{V_{RF}}{r_{be}} \cdot R_L = -\frac{\beta \cdot R_L}{r_{be}} \cdot V_{RF} \quad (\text{at resonance})$$



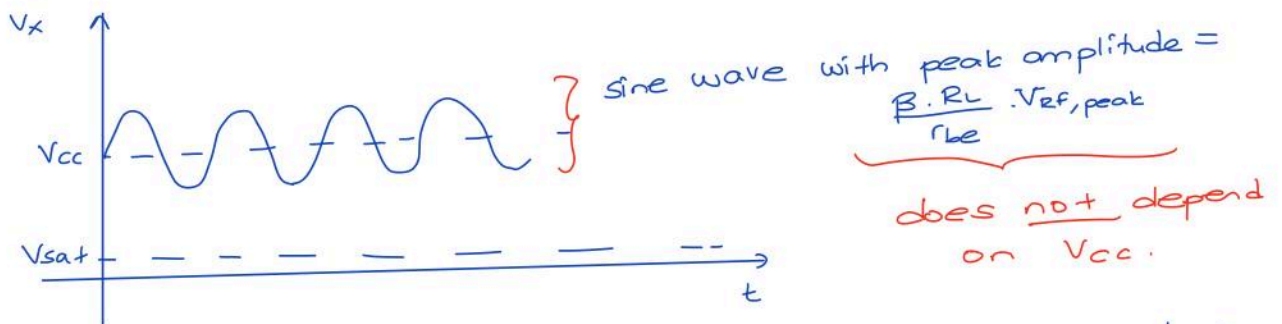
peak-to-peak voltage
 $= 2V_{CC} - V_{sat} - V_{sat}$
 $= 2(V_{CC} - V_{sat})$
 peak voltage $= V_{CC} - V_{sat}$

* If the input voltage V_{RF} sufficiently large, the magnitude of the sine wave is determined by V_{CC} .

* LC resonance only allows a sinusoid at 27 MHz at the output.



* if the input voltage V_{RF} is not sufficiently large:

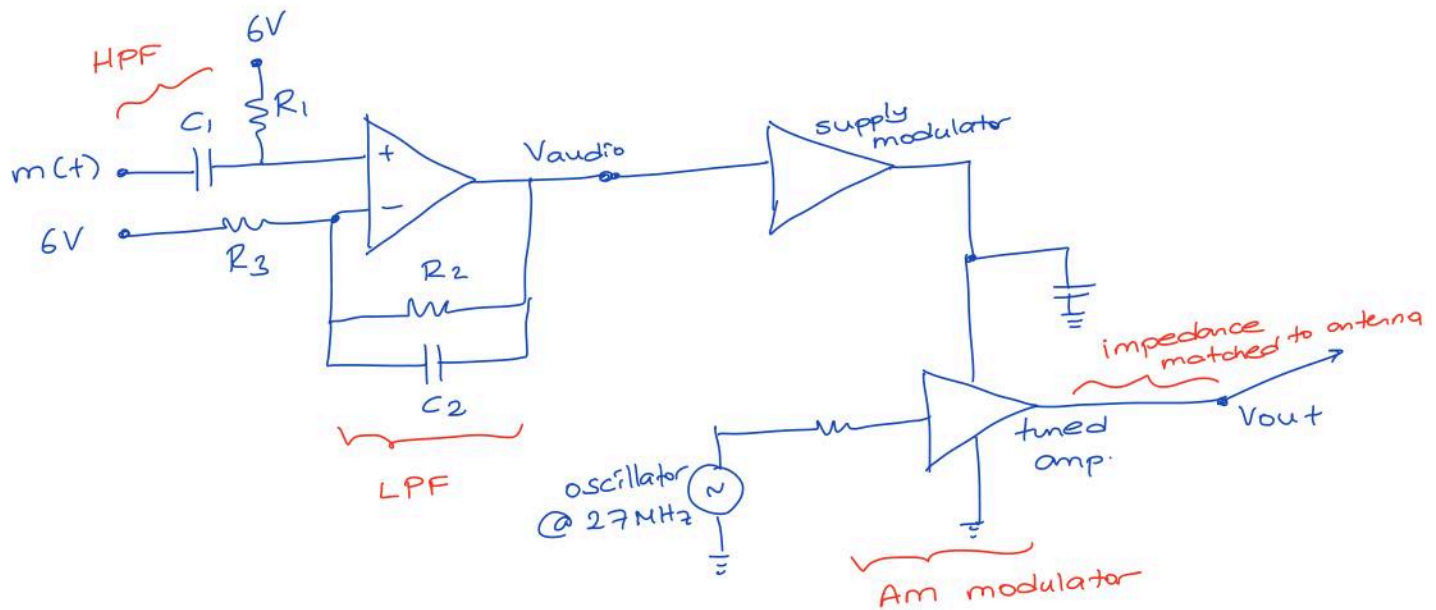


To see AM modulation, input voltage V_{RF} has to be sufficiently large.

TRC-11 Overview:

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Transmitter:



$$f_{c,1} = \frac{1}{2\pi R_1 C_1} \approx 15 \text{ Hz} \quad (\text{HPF})$$

$$f_{c,2} = \frac{1}{2\pi R_2 C_2} \approx 1.6 \text{ kHz} \quad (\text{LPF})$$

$$V_{\text{audio}} = 6 \text{ V} + A_v \cdot m(t)$$

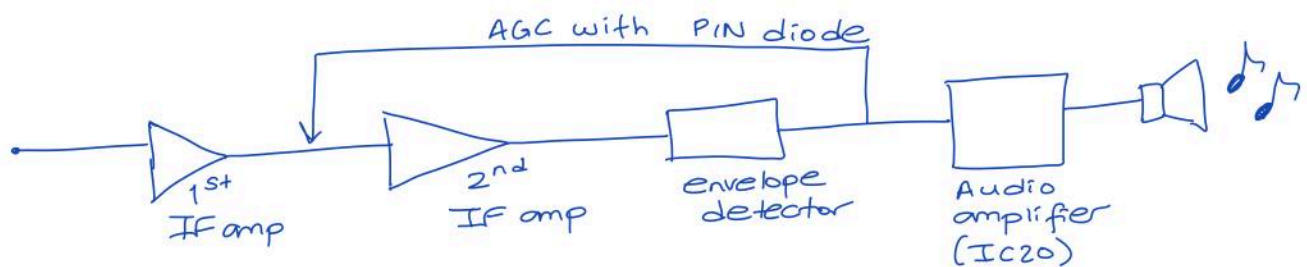
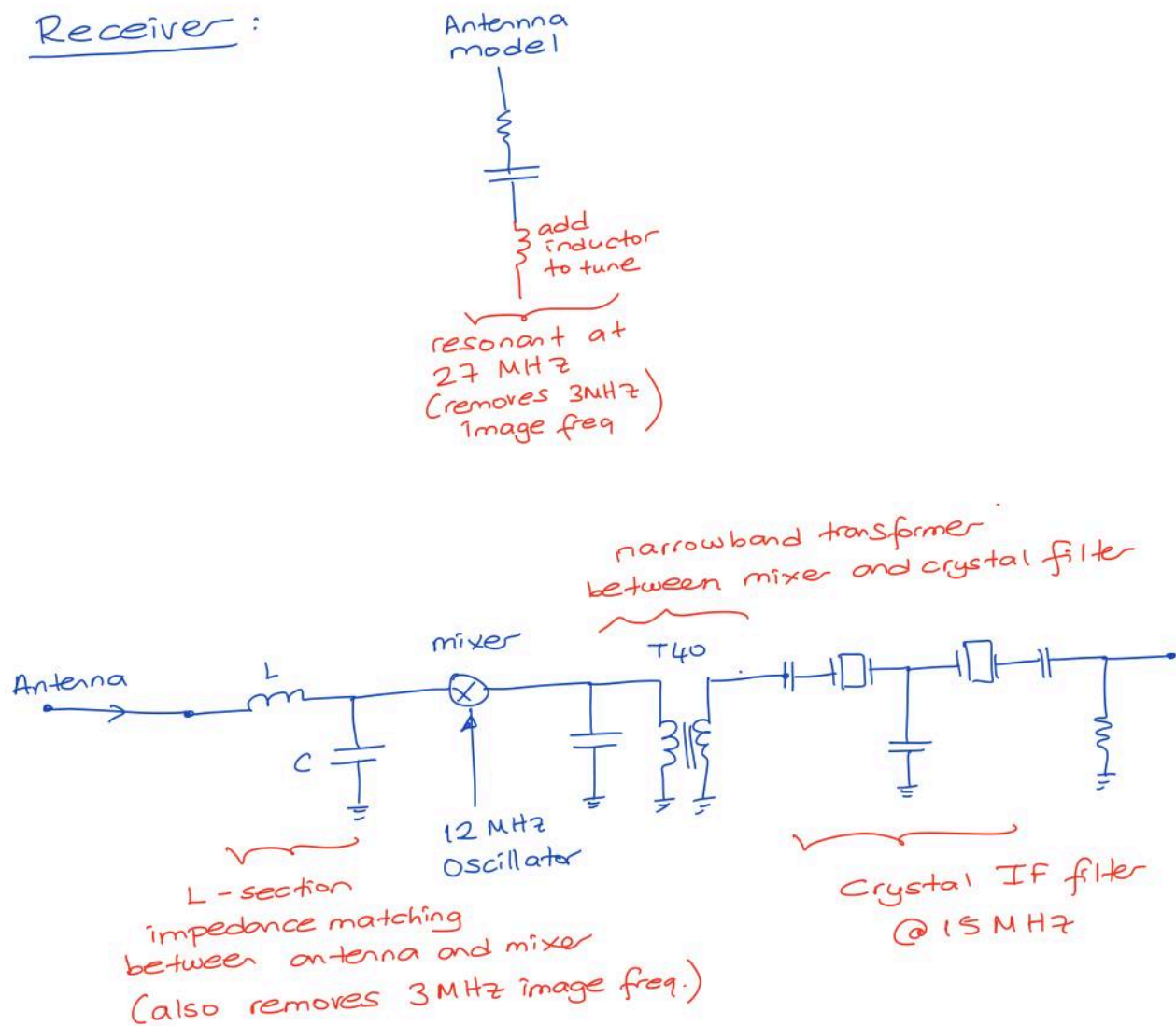
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Gain = $1 + \frac{R_2}{R_3} \approx 180$

filtered by
LPF & HPF

$$V_{\text{out}}(t) = V_c [1 + m(t)] \cdot \cos(\omega_0 t)$$

Am signal

Receiver :



What have we learned this semester?

- KCL, KVL
- Node analysis
- Capacitors, inductors
- First order circuits
- Diodes
- OPAMPs
- Phasors
- Impedance
- Transfer function
- Superposition
- Thevenin / Norton equivalent circuits
- BJT, biasing, small-signal model
- Tuned circuits, resonance
- Q factor, 3-dB bandwidth
- Filters, corner frequency
- maximum power transfer
- Transformers
- Impedance matching